Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Backend Software Engineer with 2+ years shipping **Python** and **TypeScript** services for production LLM systems, **PostgreSQL** tuning, background workers, real-time WebSockets, and **PCI-compliant** payments, plus refactoring legacy integrations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tuned **PostgreSQL** and Azure SQL schemas, indexes, and query patterns for a multi-tenant platform with row-level security, cutting dashboard latency from 1.2s to 350ms for 500+ users.
- Rebuilt LLM orchestration as fault-tolerant FastAPI **background workers** with explicit job-state transitions and failure isolation, tripling throughput across 1K+ AI jobs and sustaining 99%+ reliability for 5K+ monthly integrations.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped a production **LLM** grading platform integrating GPT-4o, Claude, and Qwen-VL with OCR, serving 10K+ daily requests for 150+ instructors with stable API latency at peak load.
- Built concurrent batch processing with timeout-aware model fallback and secure PDF, DOCX, and ZIP ingestion over **REST APIs** and AWS RDS, processing 1K+ files per batch with fault-tolerant execution.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Refactored large-scale **legacy** API integrations into modular object-oriented Singleton and Factory designs with pagination and caching, cutting technical debt 20% and improving response times 30% for low-bandwidth users.
- Led an ETL redesign on AWS Glue and S3, boosting pipeline efficiency 68%, and automated nightly Jenkins **CI/CD** with 20+ unit and integration suites across three operating systems.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **event-driven** TypeScript and Node.js microservices with real-time WebSocket events for resource allocation, billing, leads, and notifications, synchronizing 2M+ records across distributed locations with conflict resolution.
- Secured releases with JWT, role-based access control, and Snyk and OWASP dependency scanning in Jenkins CI/CD, shipping Docker builds with rolling **zero-downtime** updates and 95%+ build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Integrated **PCI-compliant** multi-currency payments across cards, UPI, wallets, and PayPal with fraud checks and idempotent webhooks for refunds, disputes, and chargebacks, cutting manual payment handling by 70%.
- Built Python Flask booking APIs with Firebase **real-time** sync and transactional locking against double-booked time slots, sustaining sub-200ms latency and 40% faster confirmations across 50K+ booking records.

Skills

Languages: TypeScript, Python, JavaScript, SQL, Java
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Webhooks, Microservices, Background Workers
Data: PostgreSQL, Query Optimization, Indexing, SQL Server, MongoDB, Firebase, Databricks, Spark
Infrastructure: AWS (S3, RDS, Glue), Azure, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Sentry
AI & Voice: LLM Integration, Claude, GPT-4o, OCR, Twilio, Prompt Chaining
Security & Practices: PCI-Compliant Payments, Row-Level Security, JWT, RBAC, Snyk, Legacy Refactoring, Agile

Projects

Real-Time Voice Agent Orchestrator | *TypeScript, Node.js, Twilio, WebSockets, Claude, PostgreSQL, AWS SQS*

- Designed a TypeScript voice-agent orchestrator bridging **Twilio** media streams over WebSockets to streaming STT, Claude, and TTS, with a Postgres call-state machine, barge-in handling, and SQS post-call PII redaction.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes, Plaid*

- Built a **fintech** platform in TypeScript and Next.js with Plaid bank linking, a token-based firewall, and WebSocket real-time updates, deployed on Kubernetes for 50K+ users and 1M+ transactions.

HMIS – Hospital Management Information System Live | *TypeScript, MERN, Twilio, ABDM APIs, Sentry*

- Won Smart India Hackathon with a TypeScript platform using consent-based ABDM records access and **Twilio** SMS reminders and confirmations, managing 100+ daily appointments and cutting administrative workload 30%.